

Toward A CONUS-Wide, Variant-Free FVS using FIA Data

A Unified Approach for National Forest Growth and Yield Modeling



Presented by:

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June 23, 2026

Introduction

We have begun modeling tree growth using Forest Inventory and Analysis (FIA) remeasurement data for all forested conditions in the continental US (CONUS).

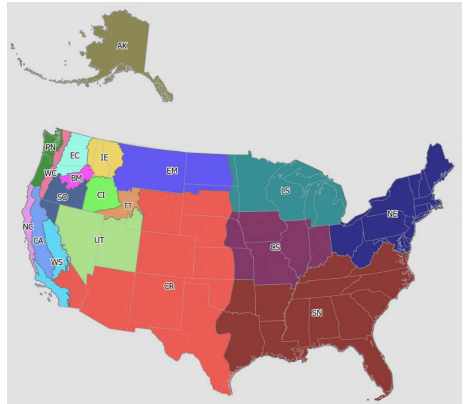
Our objective is to grow **any tree list** from plots in CONUS in a single, **variant-free** model.

Our principles:

- Use parsimonious, biologically consistent model forms for all species,
- Reduce or eliminate *site index* as a productivity measure, and
- Require no calibration when verified against FIA data.

Motivation

- FVS has around 25 variants.
- Variants differ in basic model design and function.
- Species behave differently depending on the variant.
- Some equations have not been updated to make use of new data.
- Site productivity is handled differently among variants.
- Some variants validate poorly.



Questions



Open Invitation



Anyone who is interested can join in the fun. We are setting up a Git Repository https://github.com/gregjohnsonbiometrics/fvs_remodeling to manage the project and have a private (for now) Google Drive to house the data files.

Contact one of the authors for more information!

Challenges using FIA Data for Modeling

- Multiple Condition Plots

- Insures unbiased for sample estimates (a good thing).
- Conditions can be a small area with too few trees to summarize ("plot" size).
- Very different conditions can give poor stand estimates (e.g., basal area per acre (ba), height of 40 largest trees (ht_{40}), basal area per acre in larger trees (bal)).

- Growth Periods

- 5- or 10-year (and others) intervals for same species (standardize or annualize?).
- Measurements throughout the year (may not match a growth year).

- Treatment Histories

- Identifies recent (measurement period) thinned but no description of intensity.
- No information on genetics, fertilization, and other treatments.
- Identifies recent tree damage (e.g., insects and weather) but not historic.

- Measurements

- 1-inch minimum dbh .

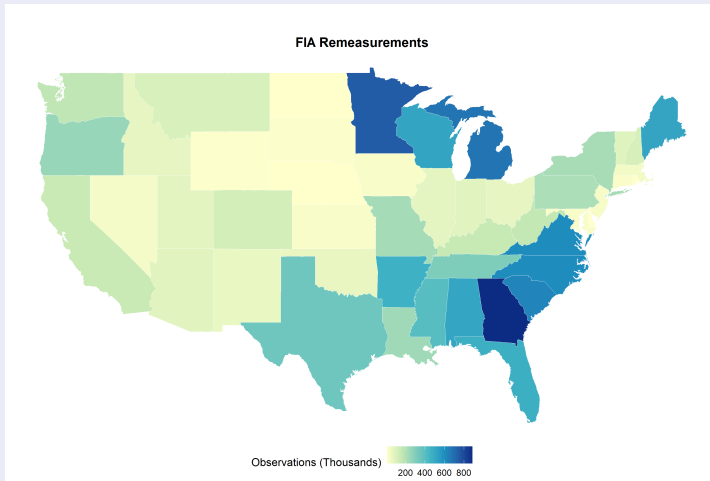
- Productivity

- Multiple definitions of site index (si) across a species range (e.g., base age).
- Site tree selection may not match site curve definition.
- Ages on older stands.

Progress

Data Retrieval and Screening

We have pulled FIA remeasurement data for CONUS. This resulted in 10870980 records from 390 species before screening for errors and missing data.



Progress

Data Retrieval and Screening

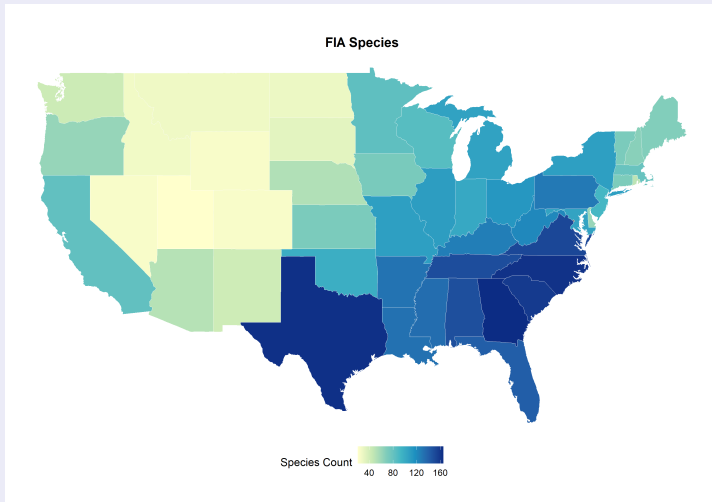


Figure 2: Map of US States scaled by FIA Species

The Site Productivity Conundrum

Site index is a mess. For example, plots with Douglas-fir have 36 site tree species using 36 site index equations, and 3 base ages.

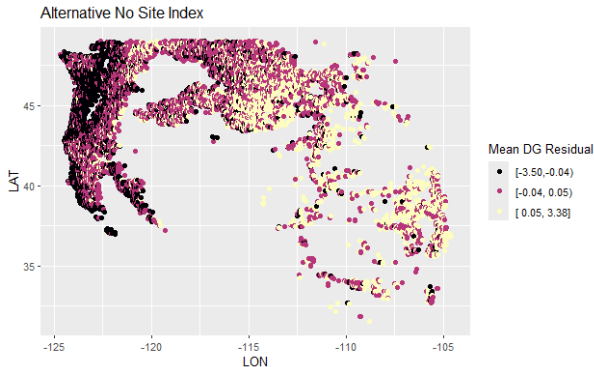
Site index can be poorly (or not) defined for young stands, unevenaged stands, or some species mixes.

We began to search for alternatives:

- ClimateSI
- Climate and environmental variables (e.g. precipitation, temperature, growing degree days)
- Embedded variables (primary and secondary principal components of climate and environment variables)
- Asymptotic Above Ground Biomass
- Remote Sensing Metrics

The Site Productivity Conundrum

A productivity (or location) variable is needed. Below are Douglas-fir diameter growth residuals from a model without a site or location effect. The points are the mean residual for an FIA condition. Darker colors are under-predictions, light colors over-predictions. Productivity gradients are apparent.

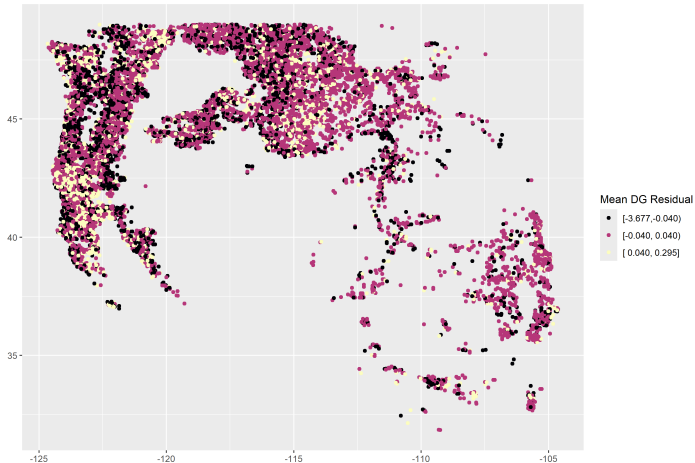


The Site Productivity Conundrum

Diameter Growth Residuals

Incorporating elevation and extreme minimum temperature does help, but it does not completely account for the spatial effects.

State - Unit - County- Plot Mean Diameter Growth Residuals

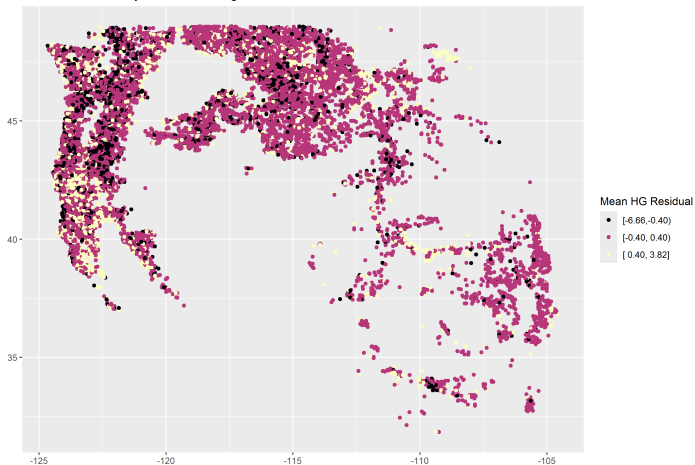


The Site Productivity Condundrum

Height Growth Residuals

Incorporating elevation, min/max temperature difference, and extreme minimum temperature does help, but as with diameter growth it does not adequately account for the spatial effects.

State - Unit - County- Plot Mean Height Growth Residuals



Growth Equation Modeling

We have preliminary equations developed for:

- Diameter Growth
- Height Growth
- Mortality

and we have in development equations for:

- Change in Crown Length
- Height imputation

Work has not begun on:

- Crown dimensions imputation
- Species mapping for species with low sample sizes

Early Results

Diameter Growth Equations

We have had good success with ORGANON's diameter growth equation¹, however, we wanted to attempt to streamline the equation and alleviate an issue where proper behavior over diameter at breast height (dbh) is problematic for many species.

Currently we have:

$$\Delta dbh = e^{(\beta_0 + \beta_1 \log(\frac{(dbh+1)^2}{(cr*ht+1)\beta_3}) + \beta_2 \frac{bal^{\beta_4}}{dbh+2.7} + \beta_5 elev + \beta_6 EMT)} \quad (1)$$

where:

- dbh = diameter at breast height (inches)
- cr = crown ratio (fraction of total height)
- ht = total height (feet)
- bal = basal area in larger trees ($feet^2/acre$)
- $elev$ = elevation above sea level (feet)
- EMT = extreme minimum temperature ($^{\circ}C$)

¹Zumrawi, A.A. and D.W. Hann. 1993. Diameter growth equations for Douglas-fir and Grand Fir in the Western Willamette Valley of Oregon. Forest Research Laboratory Research Contribution 4. Oregon State University.

Early Results

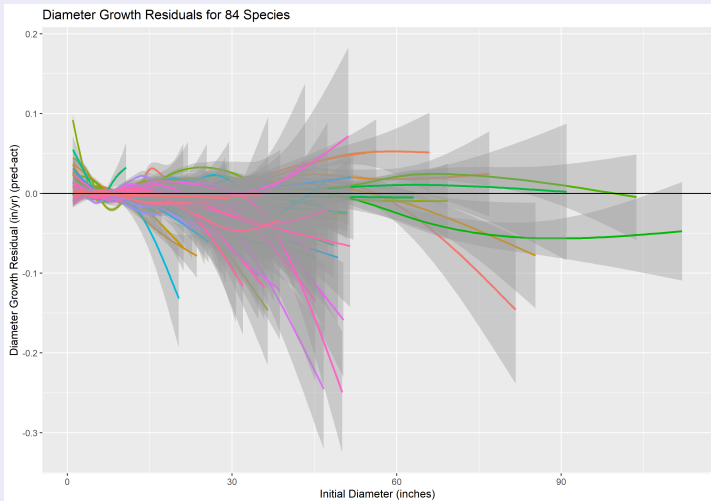
Diameter Growth Equations

We fit Equation 1 to 84 species in CONUS (limited to species with ≥ 5000 growth observations) resulting in 3481225 observations. The top ten species in terms of observations are:

FIA SPCD	Name	Observations
316	red maple	283321
131	loblolly pine	269719
318	sugar maple	170961
202	Douglas-fir	156486
12	balsam fir	150407
746	quaking aspen	149764
802	white oak	103638
241	northern white-cedar	98956
611	sweetgum	78754
122	ponderosa pine	74938

Early Results

Diameter Growth Equations



Diameter Growth Equations



Early Results

Height Growth Equations

Currently we have fit a differentiated form of the Chapman-Richards equation²:

$$\Delta ht = ht_{max} \beta_1 \beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl - \beta_8 cch^{0.5} - \beta_5 elev + \beta_6 TD^{0.5} + \beta_7 EMT)} (1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \quad (2)$$

to 3069754 growth observations, estimating β_0 through β_8 . Where:

- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- cch = crown closure at tree tip (fraction of acre)
- $elev$ = elevation above sea level (feet)
- TD = temperature difference ($^{\circ}C$)
- EMT = extreme minimum temperature ($^{\circ}C$)

ht_{max} was estimated by finding the maximum height observed in the FIA data set for each species.

²Chapman, D.G. 1961. Statistical problems in population dynamics. In: Neyman, J., ed. Proceedings of the fourth Berkeley symposium on mathematical statistics and probability. Vol. 4. Berkeley, CA: University of California Press: 153-186.

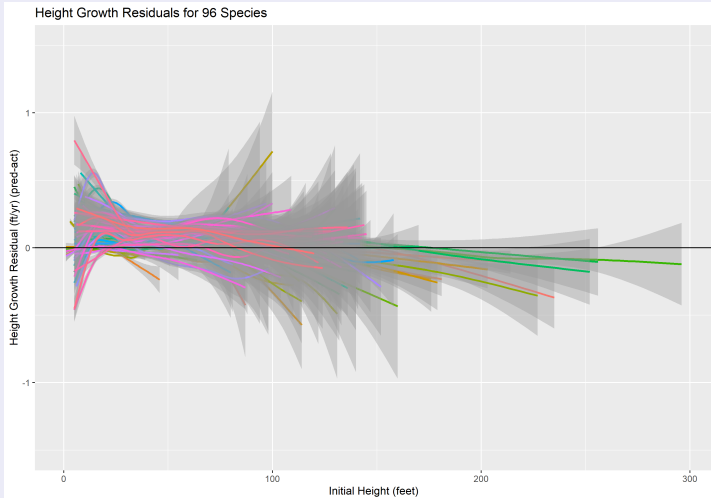
Early Results

Height Growth Equations

We fit Equation 2 to 96 species in CONUS (limited to species with ≥ 5000 growth observations) resulting in 3069754 observations.

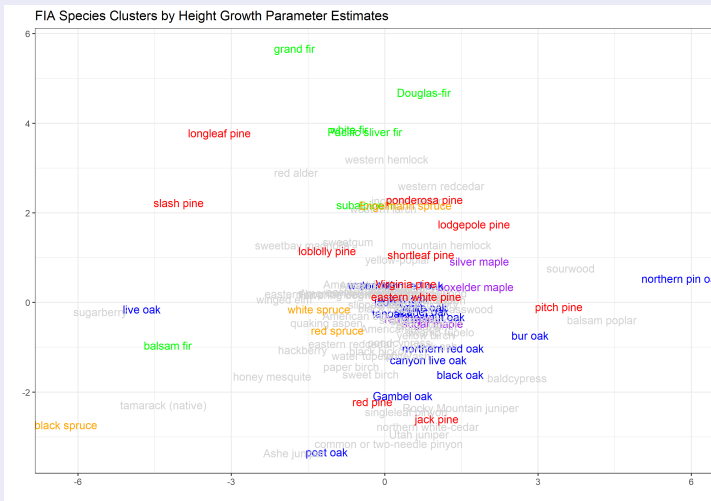
Early Results

Height Growth Equations



Early Results

Height Growth Equations



Mortality

We using a Gompit function based on crown variable to predict mortality (survival). Equation 3 contains a base-line mortality rate (β_0) that is modified by crown ratio and a crown competition measure.

Currently we have:

$$P_{surv} = 1 - e^{(-e^{(\beta_0 + \beta_1(cr + 0.01)^{\beta_2} + \beta_3 cch^{\beta_4}))})} \quad (3)$$

where:

- P_{surv} : probability of survival
- cr : crown ratio
- cch : crown closure at tree tip

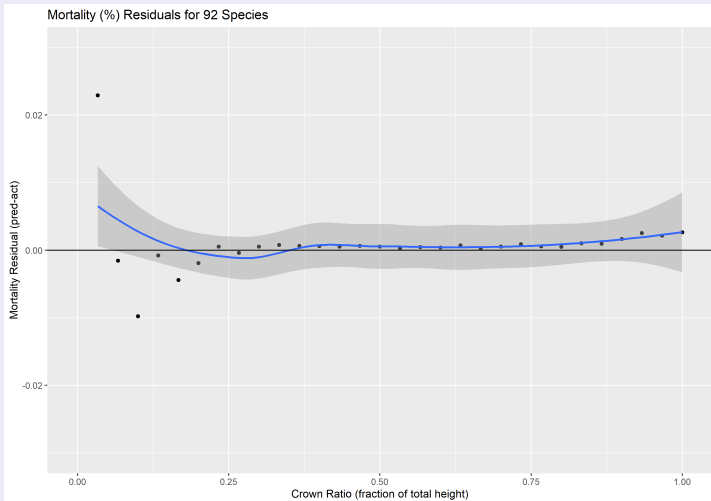
Early Results

Mortality

We fit Equation 3 to 92 species in CONUS (limited to species with ≥ 5000 growth observations) resulting in 4148232 observations.

Early Results

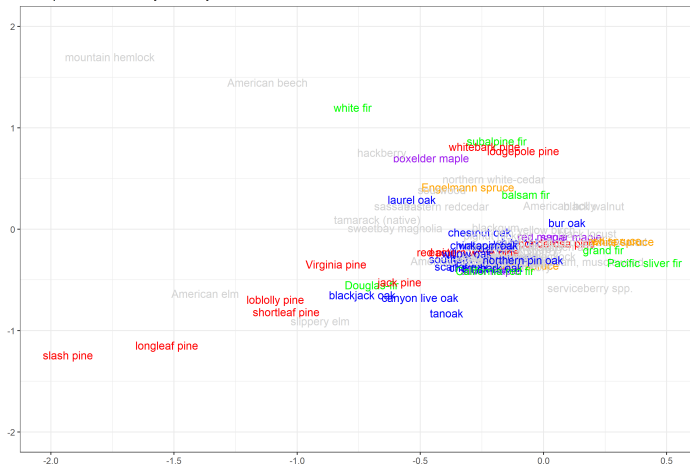
Mortality



Early Results

Mortality

FIA Species Clusters by Mortality Parameter Estimates



Early Results

Change in Crown Length

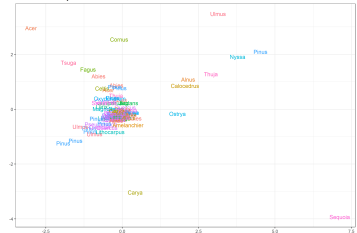
- We are in the early stages of exploring predicting change in height to live crown (Δh_{tlc}) directly.
- We have found a large number of observations where h_{tlc} decreases and preliminarily have dropped these data points.
- Height growth (Δht), crown ratio (cr), crown competition factor in larger trees ($ccf1$), and crown closure at tree tip (cch) are looking promising.

Work to be Done

- Improvement in site (location) effects
- Continue to refine height growth equation for consistency across species.
- Explore parameter clustering techniques (perhaps at the genus level) for species mapping.
- Build (or rebuild) imputation equations for:
 - Δh_{tlc}
 - ht
 - $htlc$
 - dbh ???
 - mcw (maximum crown width)
 - lcw (largest crown width)
- Build a testing simulation framework and a growth verification system to evaluate bias and precision.

Will genus clustering work?

Genus Clusters by Diameter Growth Parameter Estimates



Concurrent Work

Aaron Weiskittel is undertaking a global recalibration of existing FVS variants and attempting to build a **species-free** FVS. Equations are keyed to functional traits and climate. It covers rare and unsampled species, where per-species fits run out of data.

Concurrent Work

Six FVS builds under comparison (CONUS-wide, identical FIA stands)

Existing FVS engine

1. Default - FVS as USFS ships it

2. Recalibrated - same engine, coefficients re-fit to national FIA

New fvs-conus equations

3. Species-specific - each species its own parameters

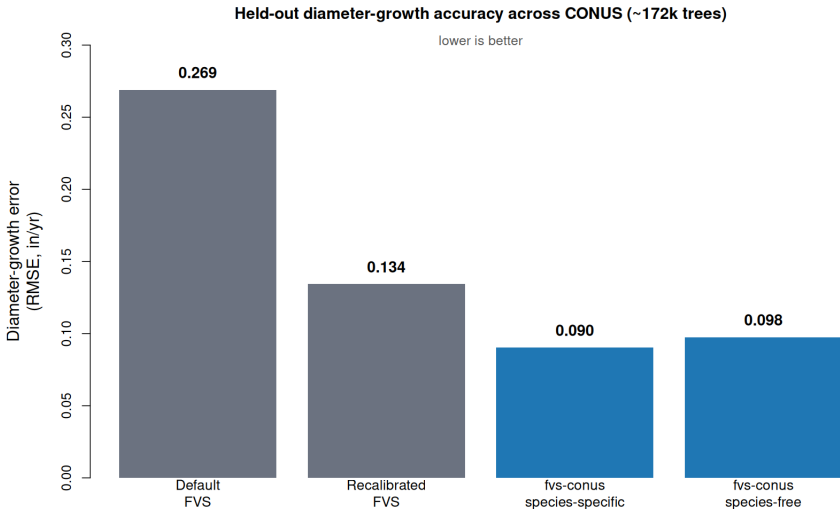
4. Species-free - species enters only via traits (one engine, any species)

Greg's ORGANON lineage

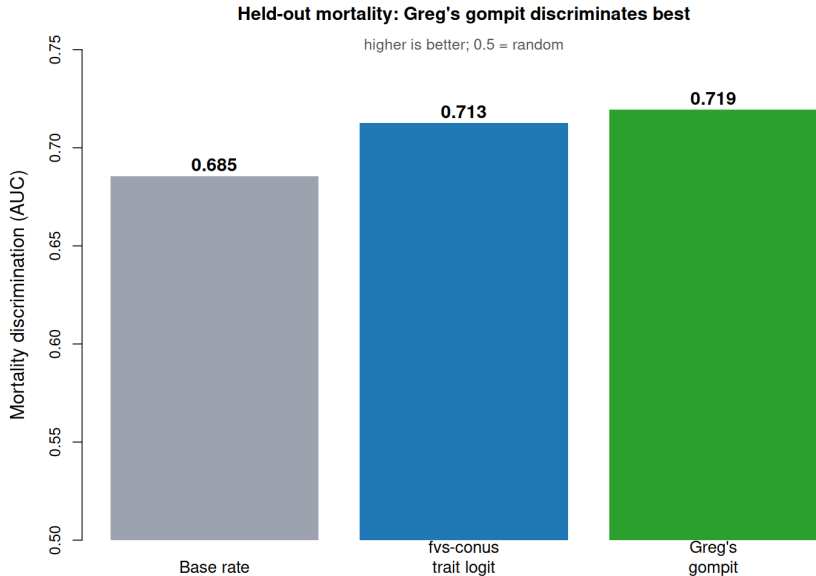
5. Gompit - fvs-conus growth + Greg's crown-based mortality

6. Full ORGANON - Hann growth + gompit mortality + ORGANON crown/ht

Concurrent Work



Concurrent Work



Thanks!

